

Basic String Concepts in Python

1. What is a String?

- A string in Python is a sequence of characters enclosed within **single** (' '), **double** (" "), or **triple quotes** (''' '' or "" "" """).
- Strings are **immutable** and **ordered**. `s1 = 'Hello'`
- Ex.
 - `s2 = "Python"`
 - `s3 = '''Multi-line String'''`

2. Indexing

- Indexing **allows accessing characters** using their position.
- **Forward Indexing:**
 - `H e l l o`
`0 1 2 3 4`
 - `print("Hello"[0]) # H`
- **Reverse Indexing:**
 - `H e l l o`
`-5 -4 -3 -2 -1`
 - `print("Hello"[-1]) # o`

3. String Slicing

- **Syntax:** `string[start:end:step]`
 - `text = "Python"`
 - `print(text[0:3]) # Pyt`
 - `print(text[2:]) # thon`
 - `print(text[:3]) # Pyt`
 - `print(text[::-1]) # nohtyP (reverse)`

4. Reverse String

- `text = "Vishal"`
 - `print(text[::-1])` # lahsiv

5. Important Notes

- Strings are **immutable** (cannot change directly):
 - `text[0] = 'J'` # **Error**

➤ Valid Way to Modify String:

- `text = 'Python'`
`text = 'J' + text[1:]`
`print(text)` # Jython
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Python String Methods

Core Methods with Examples and Cases

1. len()

Returns length of string

example:

- `print(len("hello"))` # 5
- `print(len(""))` # 0 (empty string)
- `print(len(" "))` # 2 (spaces count)

2. count()

Counts occurrences of substring

example:

- `print("banana".count("a"))` # 3
- `print("banana".count("an"))` # 2
- `print("banana".count("z"))` # 0

3. title()

Converts first letter of each word to uppercase

example:

- `print("hello world".title())` # Hello World

4. lower()

Converts all characters to lowercase

example:

- `print("HELLO".lower())` # hello

5. upper()

Converts all characters to uppercase

example:

- `print("hello".upper()) # HELLO`

6. istitle()

Returns True if each word starts with uppercase

example:

- `print("Hello World".istitle()) # True`
- `print("hello world".istitle()) # False`

7. islower()

Returns True if all characters are lowercase

example:

- `print("hello".islower()) # True`
- `print("Hello".islower()) # False`

8. isupper()

Returns True if all characters are uppercase

example:

- `print("HELLO".isupper()) # True`
- `print("Hello".isupper()) # False`

9. find()

Finds first occurrence index, returns -1 if not found

example:

- `print("banana".find("a")) # 1`
- `print("banana".find("z")) # -1`

10. rfind()

Finds last occurrence index

example:

- `print("banana".rfind("a")) # 5`

11. index()

Similar to find(), raises error if not found

example:

- `print("banana".index("a")) # 1`
- `# print("banana".index("z")) # Error`

12. rindex()

Finds last occurrence index, raises error

example:

- `print("banana".rindex("a")) # 5`

Feature	find()	index()
Return value if substring is found	Returns index	Returns index
Return value if substring is not found	Returns -1	Raises error (ValueError)
Use case	Safer to use when unsure about substring	Use when substring is guaranteed to exist
Syntax	<code>string.find(substring)</code>	<code>string.index(substring)</code>

13. replace()

Replace substring in string

example:

- `print("hello world".replace("world", "Python")) # hello Python`

14. isalpha()

True if all chars are alphabets

example:

- `print("abc".isalpha()) # True`
- `print("abc123".isalpha()) # False`

15. isalnum()

True if alphanumeric only

example:

- `print("abc123".isalnum()) # True`

- `print("abc 123".isalnum()) # False`

16. `isdecimal()`

Checks only decimal characters

example:

- `print("123".isdecimal()) # True`
- `print("12.3".isdecimal()) # False`

17. `isdigit()`

True if all characters are digits

example:

- `print("123".isdigit()) # True`
- `print("123".isdigit()) # True (superscript 3)`

18. `strip()`

Removes leading and trailing spaces

Cases:

- `print(" hello ".strip()) # hello`

19. `lstrip()`

Removes only left-side spaces

Cases:

- `print(" hello".lstrip()) # hello`

20. `rstrip()`

Removes only right-side spaces

Cases:

- `print("hello ".rstrip()) # hello`

21. `split()`

Splits string into list by given substring default by space

Cases:

- `print("a,b,c".split(",")) # ["a", "b", "c"]`